N83 D18 shared-bit encoding

Construction and semantic validation · frozen protocol dated 2026-10-20

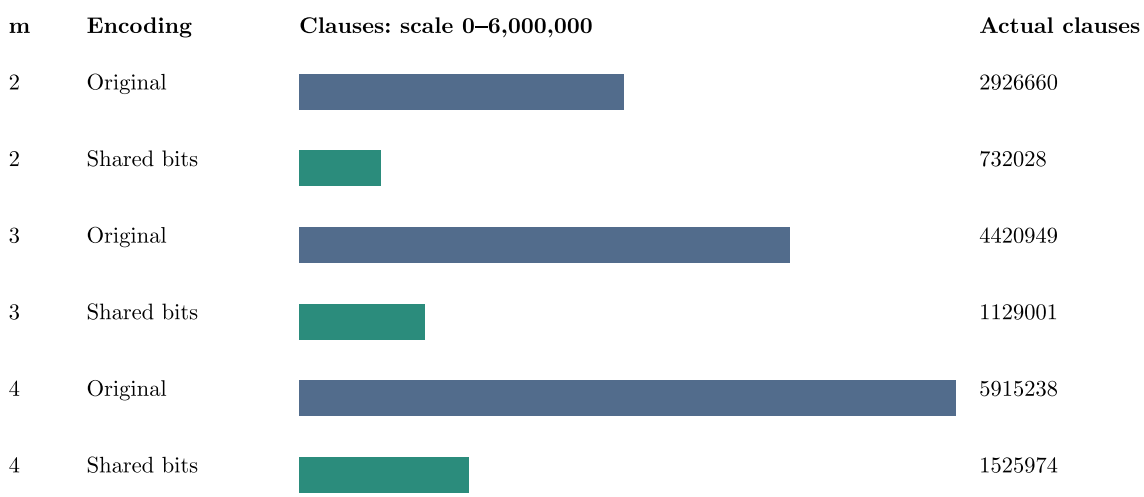
Requested scope and result boundary

The user requested completion of PR 2271 and continued D18 shared-bit encoding optimization. This report covers exact construction counts for the full retained 16,627-class D18 base at m=2,3,4, Boolean equivalence, 23 passing example tests and the implemented common-public m=4 comparison. The complete release-library suite and paired solver execution are pending. The source baseline is `e294f8c927b019f59e6552039d32e6b4ba237f84`. It is the encoding baseline; a complete verified N83 IC cost is still an open gate of the broader factor-base study.

D18: actual SAT construction counts

N83 · 16,627 fixed representatives · both encodings constructed for m=2,3,4

66 common root bits; accepted domain and choice wires unchanged.



m	Variables, both encodings	XOR rows, both encodings
2	90258	1743
3	145928	2822
4	201598	3901

One deterministic construction per cell. Exact integer counts; no sampling interval. Source: `construction.json`, native source digests recorded in that receipt. No solver search or runtime comparison is included in this construction check.

Exact transformation

For each bit that agrees across all K representatives, replace K selector implications by one common-bit unit. At least one selector is active, so the unit follows from the original implications. Conversely, that unit satisfies every removed implication. Varying bits retain their original clauses. The replacement covers shared ones and shared zeros.

Variables, selector order, phase wires and all other CNF/XOR constraints are preserved. Units are installed after domain construction because the solver eagerly simplifies later clauses against root assignments. The default API retains the original encoding; the optimization is explicit.

D18 describes the 18-dimensional F2-linear source space from which the retained subgroup representatives were selected. The working field stays F2^83. Shared bits constrain each selected root before Frobenius; all 83 phases and both point signs remain available in the factor-base closure.

For m domains, c common bits and K representatives, the exact removed clause count is `m*c*(K-1)`. The receipt records actual allocations and checks that each matches its preflight. This is an encoding-size result. Runtime improvement and complete IC cost are not measured by this check.

Identity and reproducibility

Curve: `icv1-f2m83-tm6151469093347-debefd74`, a=0, cofactor 4, subgroup order 241785163923079621668568 9, polynomial z^83+z^45+z^2+z+1. The retained representative CSV is pinned by SHA-256: 534e58ced979e05ec747eb462aa1d63135d8004813d094c522a4d7160c03c4a4.

Full D object BLAKE3: e0c24980207cbb15fa57ec76bcfc6a628a763a80834cba1117ca83ea04678682. It contains 2,760,082 signed/phase-labeled points. The earlier export and independent subgroup/label replay remain the point-object evidence; this construction check reads its selected source-coordinate inventory.

Actual shared bits: 66. Ordered representative BLAKE3: f9fb7d25682a29556c133d74690b447a3ed5f125b6248dd89b0089be1ad4aced.

The native tool builds six models with target x fixed to public corpus index zero. It compares preflight against actual counts and checks that choice-wire allocation is identical between encodings. It performs no SAT search. The receipt records SHA-256 of the library module and tool. Host, commands, exits and validation logs are preserved beside the receipt. This local arm64 construction is a structural check on a shared host (L0). Its memory was not capped by a container. The separate 4 GiB Linux construction gate remains required before the paired card can launch.

Proof and executable checks

Core Lean 4.34.1 checks three theorems in `lean/SharedRootBits.lean`: factorization, equivalence within the unchanged context, and the exact natural-number count identity. The first two have no axioms; the count theorem uses only `propext`. There is no admitted theorem or native evaluation. `PROOF_MAP.md` gives each exact statement and the Rust bridge. The Rust compiler and the full SAT solver are outside that formal proof. The additional `lean4checker` implementation is unavailable on this host.

The count polynomial also has native certificate IDC1hbc6bfea0a569542b, issued on 32 points and checked on 32 additional points. That fixed-field certificate has a narrower scope than the Lean natural-number theorem. Twenty-three affected-example tests pass, including receipt tampering and command-card order/failure handling. The implemented library tests cover exhaustive degree-3 representative/phase/x domains, shared ones and zeros, atomic caps and invalid domains. Their execution remains pending in the full release-library suite. Counts, commands and failure receipts are tracked in `README.md` and the versioned validation archives. The first example build failed with E0282; an explicit count-tuple type fixed it before the passing run. Earlier SSD-space and interrupted-build receipts are preserved in validation v1; the passing gates and compile failure are in validation v2. Neither archive claims a full-library pass.

Remaining experiment and original gates

The implemented card compares the same D object under both encodings: five A/A constructed pairs, five D A/B pairs, one U A/B pair, each with fresh general-field replay, then eighteen natural m=4 attempts across all eight public points. It stops before natural work if a control fails. A committed host card must bind the final source, binary, inputs and wrappers before launch. Its execution status is recorded separately from this report.

Natural useful rank, m=2/m=3 SAT policies beyond the completed finite events, complete cold and reused cost, linear algebra, matched same-point rho, fastest-external comparison, and best-base selection remain open. Capped queries retain their censored status. Candidate and selected-best-runtime fields remain null.

Canonical runtime dashboards and benchmark rows are unchanged: this result records exact clause counts, without a time measurement or pipeline ranking. The study graph is generated directly from `construction.json`. Sources: `PROTOCOL.md`, `PROOF_MAP.md`, `construction.json`, `README.md`, and `problems/n83-prebuilt-base-quality-m2-m4.md`.